

MSC FINANCE & BIG DATA

Optimization Methods with Applications to Finance • 2627_FI_5_3_073_E

Nonlinear Optimization

Gradient and Newton methods

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Session 4 • Tutorial + Workshop • 2 × 180 min

The problem, and how far we will go

$$\begin{array}{ll} \underset{x \in \mathbb{R}^n}{\text{minimize}} & f(x) \\ \text{subject to} & c(x) = 0 \quad \text{equality} \\ & d(x) \geq 0 \quad \text{inequality} \end{array}$$

All functions are assumed C^2 — continuous with continuous first and second derivatives — but they need not be linear.

This chapter

1. **Unconstrained optimization** — the bulk of the session

$$\min_{x \in \mathbb{R}^n} f(x)$$

2. **Gradient and Newton methods** — how a machine finds it
3. **One equality constraint** — just enough Lagrange to reach the mean-variance portfolio

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{s. t.} \quad c(x) = 0$$

The general inequality case — KKT, complementary slackness — is named at the end and left there. It is not needed for what follows.

Where we are in the fil rouge

Fil rouge — NEOMAMA Brokerage

Chapter 2 gave the client \$50,000 in municipal bonds, \$175,000 in Palmer Drugs and \$25,000 in nursing homes — a return of 8.12%. The LP had no way of knowing that Palmer and the nursing home stock move together.

Chapter	Risk is...	Model class
2	a linear index, t^tsw	LP — solved at a vertex
3	a quadratic form, $t^t w \Sigma w$	QP — solved by calculus
4	one objective among several	Multi-objective
5	a property of the whole distribution	Utility, dominance

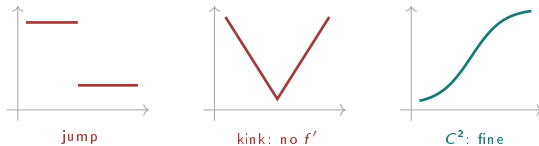
Takeaway

By the end of the afternoon you will have re-solved the client's mandate with a covariance matrix, and measured exactly what the LP answer cost her. **The preview: 15% less risk for the same return.**

Smoothness and Taylor expansion

What “ f is C^2 ” buys us

Everything in this chapter rests on being able to *approximate f near a point by a polynomial*. That needs f continuous and twice differentiable.



Taylor expansion in one variable

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2 + \theta((x - x_0)^2), \quad x \rightarrow x_0.$$

Read it as: *near x_0 , every smooth function is a parabola* $f(x) \approx \gamma + \alpha x + \beta x^2$. All of unconstrained optimization is the study of that parabola.

Two Taylor examples

A polynomial is its own expansion

$f(x) = 3x^2 + x + 1$, at $x_0 = 0$:

$$f(0) = 1, \quad f'(0) = 1, \quad f''(0) = 6.$$

$$f(x) = 1 + x + \frac{1}{2}(6)x^2 = 1 + x + 3x^2. \checkmark$$

Exact, with no remainder: a quadratic *is* its own second-order Taylor polynomial, around any x_0 .

A transcendental function is not

$f(x) = e^x$, at $x_0 = 0$: all derivatives equal 1, so

$$e^x \approx 1 + x + \frac{1}{2}x^2,$$

the first three terms of $e^x = \sum_{i=0}^{\infty} \frac{x^i}{i!}$.

At $x_0 = 5$: $e^x \approx e^5 + (x - 5)e^5 + \frac{1}{2}(x - 5)^2e^5$. The approximation is local — it is only good *near* the point you expanded about.

Takeaway

Newton's method, later in this chapter, is nothing but: *replace f by its Taylor parabola, jump to the bottom of the parabola, re-expand, repeat*. Everything else is bookkeeping.

Optimality in one variable

Suppose $f'(x_0) = 0$. Then the expansion loses its linear term:

$$f(x) = f(x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2 + \theta(\cdot).$$

If $f''(x_0) > 0$, the second term is strictly positive for $x \neq x_0$, so

$$f(x) > f(x_0)$$

for every x close enough: x_0 is a **local minimum**.

If $f''(x_0) < 0$, the same argument with the sign reversed gives a **local maximum**.

One variable, summarised

Necessary	$f'(x_0) = 0$ and $f''(x_0) \geq 0$
Sufficient	$f'(x_0) = 0$ and $f''(x_0) > 0$

Where it breaks

The gap between $\geq$ and $>$ is real. For $f(x) = x^3$ at $x_0 = 0$ we have $f' = f'' = 0$, and x_0 is neither a maximum nor a minimum. Second-order information is not always enough.

Gradient and Hessian

The gradient

Definition 1 (Gradient)

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(x) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x) \end{pmatrix}.$$

It points in the direction of *steepest increase*, and its negative $-\nabla f$ in the direction of steepest decrease. That single fact is the entire gradient method.

Worked example

$$f(x) = 3(x_1)^2(x_2)^3 + (x_2)^2(x_3)^3$$

$$\nabla f(x) = \begin{pmatrix} 6x_1x_2^3 \\ 9x_1^2x_2^2 + 2x_2x_3^3 \\ 3x_2^2x_3^2 \end{pmatrix}$$

Differentiate one variable at a time, holding the others fixed. The middle entry collects contributions from *both* terms because x_2 appears in each.

The Hessian

Definition 2 (Hessian)

$$H_f(x) = \nabla^2 f(x) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(x) \right)_{i,j}$$

For $f \in C^2$ the mixed partials commute ($\partial^2 f / \partial x_i \partial x_j = \partial^2 f / \partial x_j \partial x_i$), so H_f is **symmetric**. We will use that constantly.

Symmetry is why “positive definite” is even a meaningful test, and why H_f^{-1} behaves in the Newton step.

The example we keep coming back to

$$f(x_1, x_2) = x_1^2 + x_2^2$$

$$\nabla f(x) = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix}, \quad H_f(x) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

And one with a cross term

$$f(x, y) = x^2 + xy + 2y^2 + 3$$

$$\nabla f = \begin{pmatrix} 2x + y \\ x + 4y \end{pmatrix}, \quad H_f = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}.$$

Here H_f is constant — true for every quadratic, and false in general.

Taylor in n dimensions

Second-order expansion

$$f(x) = f(x_0) + {}^t\nabla f(x_0)(x - x_0) + \frac{1}{2} {}^t(x - x_0)H_f(x_0)(x - x_0) + \theta(\|x - x_0\|^2)$$

Check it on $f(x) = x_1^2 + x_2^2$ at $x_0 = (0, 0)$

$$f(x) = \underbrace{0}_{f(x_0)} + \underbrace{(0, 0) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}}_{\text{gradient term}} + \frac{1}{2}(x_1, x_2) \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \frac{1}{2}(2x_1^2 + 2x_2^2) = x_1^2 + x_2^2. \checkmark$$

Takeaway

Compare with Chapter 1: the quadratic form ${}^t z Q z$ is back, with $Q = H_f(x_0)$ and $z = x - x_0$. Whether x_0 is a minimum is therefore a question about the *sign* of a quadratic form — which is the next slide.

Positive semidefinite and positive definite

Definition 3

Let A be symmetric $n \times n$.

$$A \succeq 0 \iff {}^t z A z \geq 0 \quad \forall z \quad \text{PSD}$$

$$A \succ 0 \iff {}^t z A z > 0 \quad \forall z \neq 0 \quad \text{PD}$$

With $A = H_f(x_0)$ and $z = x - x_0$, this is exactly the quantity that decides the sign of $f(x) - f(x_0)$ near a stationary point.

The next slide turns the definition into a test you can actually run — three determinants and a table.

Why this matters for portfolios

A covariance matrix is positive semidefinite *by construction*:

$${}^t w \Sigma w = \text{Var}({}^t w \tilde{r}) \geq 0 \quad \text{for every } w.$$

A variance cannot be negative, so no portfolio can make the quadratic form negative.

Takeaway

Σ fails to be *strictly* positive definite only when some combination of assets is riskless — that is, when one asset is an exact linear combination of the others. Then ${}^t w \Sigma w$ is convex but not *strictly* convex, and by Chapter 1 the minimum-variance portfolio need no longer be unique.

Testing the Hessian: minimum, maximum, or saddle?



$\nabla^2 f \succ 0$ — **positive definite**

curves up everywhere $\Rightarrow$ **MINIMUM**



$\nabla^2 f \prec 0$ — **negative definite**

curves down everywhere $\Rightarrow$ **MAXIMUM**



indefinite

up one way, down another $\Rightarrow$ **SADDLE**

The test. D_k = determinant of the top-left $k \times k$ block of $\nabla^2 f$ — the k -th *leading principal minor*.

Pattern of $D_1, D_2, \dots, D_n$ Verdict

all positive	$+$ $+$ $+$ $\dots$	minimum
alternating	$-$ $+$ $-$ $\dots$	maximum
any other pattern, $D_n \neq 0$		saddle
$D_n = 0$		test fails

In two variables that is all you need. For $\nabla^2 f = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$:

$D_1 = a$, $D_2 = \det \nabla^2 f = ac - b^2$, and

$D_2 > 0$ and $a > 0 \Rightarrow$ **minimum**

$D_2 > 0$ and $a < 0 \Rightarrow$ **maximum**

$D_2 < 0 \Rightarrow$ saddle

$D_2 = 0 \Rightarrow$ no conclusion

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Three Hessians, three verdicts

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix} \quad B = \begin{pmatrix} -2 & -1 \\ -1 & -4 \end{pmatrix} \quad C = \begin{pmatrix} 0 & -3 \\ -3 & 0 \end{pmatrix}$$

	$D_1 = a$	$D_2 = ac - b^2$	Verdict
A	$2 > 0$	$8 - 1 = 7 > 0$	minimum
B	$-2 < 0$	$8 - 1 = 7 > 0$	maximum
C	0	$0 - 9 = -9 < 0$	saddle

A Hessian of $f = x^2 + xy + 2y^2 + 3$, everywhere

B Hessian of $f = -x^2 - xy - 2y^2$, everywhere

C Hessian of $f = x^3 + y^3 - 3xy$ at $(0, 0)$

Order of work: solve $\nabla f = 0$ first, then evaluate D_1, D_2 at each stationary point.

Unconstrained optimization

Optimality conditions

Definition 4 (Stationary point)

x_0 is **stationary** if $\nabla f(x_0) = 0$.

First order — necessary

If x_0 is a local minimum then $\nabla f(x_0) = 0$.

Second order — necessary

... and $H_f(x_0) \succeq 0$.

Second order — sufficient

If $\nabla f(x_0) = 0$ and $H_f(x_0) \succ 0$, then x_0 is a strict local minimum.

Why. With $\nabla f(x_0) = 0$ the expansion reduces to

$$f(x) - f(x_0) = \frac{1}{2} {}^t(x - x_0)H_f(x_0)(x - x_0) + \theta(\cdot),$$

so the sign of $f(x) - f(x_0)$ near x_0 is the sign of the quadratic form.

$H_f(x_0) \succ 0$	strict local minimum
$-H_f(x_0) \succ 0$	strict local maximum
$H_f(x_0)$ indefinite	saddle point
$H_f(x_0)$ singular	test inconclusive

Takeaway

Chapter 1 closes the loop: if f is convex then $H_f \succeq 0$ everywhere, and every stationary point is a **global** minimum. No further checking needed.

Worked example 1 — a well-behaved quadratic

$$f(x, y) = x^2 + xy + 2y^2 + 3$$

First-order conditions.

$$\nabla f(x, y) = \begin{pmatrix} 2x + y \\ x + 4y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies \begin{cases} 2x + y = 0 \\ x + 4y = 0 \end{cases} \implies x = 0, y = 0.$$

Second-order conditions.

$$H_f = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}, \quad |H_1| = 2 > 0, \quad |H_2| = \det H_f = 7 > 0.$$

$H_f \succ 0$, so $(0, 0)$ is a strict local minimum with $f(0, 0) = 3$.

Takeaway

H_f is constant here, so it is positive definite *everywhere*: f is convex, and by Chapter 1 the local minimum is the **global** minimum. Quadratics are the easy case — and, conveniently, the case portfolio variance falls into.

Worked example 2 — a saddle and a minimum

$$f(x, y) = x^3 + y^3 - 3xy$$

(Resit 2026, Problem 4)

$$\nabla f = \begin{pmatrix} 3x^2 - 3y \\ 3y^2 - 3x \end{pmatrix} = 0 \implies y = x^2, \quad x = y^2 \implies x^4 = x \implies x(x^3 - 1) = 0.$$

Two stationary points: $(0, 0)$ and $(1, 1)$. The Hessian is $H_f = \begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix}$.

At $(0, 0)$

$$H_f = \begin{pmatrix} 0 & -3 \\ -3 & 0 \end{pmatrix}, \quad \det H_f = -9 < 0.$$

Indefinite $\Rightarrow$ **saddle point**. Along $y = x$ the function falls; along $y = -x$ it rises.

At $(1, 1)$

$$H_f = \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}, \quad |H_1| = 6 > 0, \quad \det H_f = 27 > 0.$$

Positive definite $\Rightarrow$ **strict local minimum**, $f(1, 1) = -1$.

Note what a solver would do here. Started near the origin, a naive descent method can stall at the saddle, where $\nabla f = 0$. This function is *not* convex — Chapter 1's guarantee does not apply, and neither does any promise of global optimality.

Worked example 3 — gradient and Hessian at a point

$$f(x_1, x_2) = (x_1 + x_2^2)^2 \text{ at } (1, 2)$$

(Final Exam 2025, Problem 3)

Write $s = x_1 + x_2^2$, so $f = s^2$. At $(1, 2)$: $s = 1 + 4 = 5$.

Gradient (chain rule):

$$\frac{\partial f}{\partial x_1} = 2s, \quad \frac{\partial f}{\partial x_2} = 2s \cdot 2x_2 = 4x_2s \quad \implies \quad \nabla f(1, 2) = \begin{pmatrix} 10 \\ 40 \end{pmatrix}.$$

Hessian:

$$\begin{aligned} \frac{\partial^2 f}{\partial x_1^2} &= 2, & \frac{\partial^2 f}{\partial x_1 \partial x_2} &= 4x_2, & \frac{\partial^2 f}{\partial x_2^2} &= 4s + 8x_2^2, \\ H_f(1, 2) &= \begin{pmatrix} 2 & 8 \\ 8 & 52 \end{pmatrix}, & |H_1| &= 2 > 0, & \det H_f &= 104 - 64 = 40 > 0. \end{aligned}$$

$\nabla f(1, 2) \neq 0$, so $(1, 2)$ is *not* a stationary point — the Hessian test says nothing about optimality here. The question asked only for the two objects; do not over-answer it.

Audit the machine — a correct computation, a wrong conclusion

Asked to analyse $f(x_1, x_2) = (x_1 + x_2^2)^2$ at the point $(1, 2)$:

AI assistant — response

The gradient is $\nabla f = (2s, 4x_2s)$ with $s = x_1 + x_2^2$, and the Hessian is

$$H_f = \begin{pmatrix} 2 & 4x_2 \\ 4x_2 & 4s + 8x_2^2 \end{pmatrix}.$$

At $(1, 2)$ we have $s = 5$, so

$$H_f(1, 2) = \begin{pmatrix} 2 & 8 \\ 8 & 52 \end{pmatrix}, \quad D_1 = 2 > 0, \quad D_2 = 40 > 0.$$

The Hessian is positive definite, therefore $(1, 2)$ is a **local minimum** of f .

Your audit — 15 minutes

1. Check every number. Are the gradient, the Hessian and the two minors correct?
2. The conclusion is still false. Which *hypothesis* of the second-order test was never verified?
3. Evaluate $\nabla f(1, 2)$ and $f(1, 2)$. Then find a point where f is strictly smaller.
4. State the correct sentence the assistant should have written.

Takeaway

Every arithmetic step is right. The failure is a missing *precondition* — the hardest kind of error to see, and the kind that only theory catches.

Descent methods

The shape of every algorithm in this chapter

Iterative descent

$$x_{k+1} = x_k + \alpha_k d_k$$

where $d_k \in \mathbb{R}^n$ is a **search direction** and $\alpha_k > 0$ a **step size**.

d_k is a *descent direction* if $\nabla f(x_k)^\top d_k < 0$: moving that way decreases f , at least initially.

Method	Direction d_k	Uses
Gradient (steepest) descent	$-\nabla f(x_k)$	first derivatives
Newton	$-H_f(x_k)^{-1} \nabla f(x_k)$	first and second derivatives
Quasi-Newton (BFGS)	$-B_k^{-1} \nabla f(x_k)$, $B_k \approx H_f$	first derivatives only

The step α_k is found by *line search*: minimize $\varphi(\alpha) = f(x_k + \alpha d_k)$ over $\alpha > 0$. For a quadratic this is a one-variable calculus problem with a closed-form answer, which is what we use below.

Gradient descent, step by step

$f(x) = 2x_1^2 + x_2^2$, starting from $x_0 = (1, 1)$

$\nabla f(x) = (4x_1, 2x_2)$, so $d_0 = -\nabla f(x_0) = (-4, -2)$.

Exact line search. Along the ray $x(\alpha) = (1 - 4\alpha, 1 - 2\alpha)$,

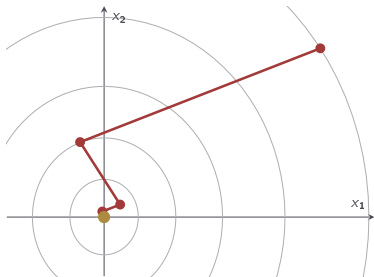
$$\varphi(\alpha) = 2(1 - 4\alpha)^2 + (1 - 2\alpha)^2 = 3 - 20\alpha + 36\alpha^2, \quad \varphi'(\alpha) = -20 + 72\alpha = 0 \implies \alpha_0 = \frac{5}{18}.$$

$$x_1 = \left(-\frac{1}{9}, \frac{4}{9}\right), \quad f(x_1) = \frac{2}{9} \approx 0.2222.$$

k	α_k	x_k	$f(x_k)$
0	—	$(1, 1)$	3.000000
1	0.2778	$(-0.1111, 0.4444)$	0.222222
2	0.4167	$(0.0741, 0.0741)$	0.016461
3	0.2778	$(-0.0082, 0.0329)$	0.001219

Each step cuts f by a constant factor of about 13.5. Steady, and never finished — the iterates approach 0 but never arrive.

Why gradient descent zig-zags



With exact line search, consecutive directions are **orthogonal**:

$${}^t\nabla f(x_{k+1}) \nabla f(x_k) = 0.$$

The iterates therefore turn a right angle at every step — the characteristic zig-zag.

The severity depends on the **condition number** $\kappa = \lambda_{\max}/\lambda_{\min}$ of the Hessian. Here $H_f = \text{diag}(4, 2)$, so $\kappa = 2$: gentle. A covariance matrix with a near-riskless combination can have κ in the thousands.

Where it breaks

Gradient descent is cheap per step and slow to converge. It uses only the *slope*, and ignores the *curvature* that would tell it how far to go.

Newton's method

Newton's method: minimize the Taylor parabola

Replace f near x_k by its second-order expansion:

$$m_k(x) = f(x_k) + \nabla f(x_k)(x - x_k) + \frac{1}{2} (x - x_k)^T H_f(x_k) (x - x_k).$$

Minimize the *model*, not the function. Setting $\nabla m_k(x) = 0$:

$$\nabla f(x_k) + H_f(x_k)(x - x_k) = 0 \quad \implies \quad x_{k+1} = x_k - H_f(x_k)^{-1} \nabla f(x_k)$$

Takeaway

In practice, never invert. Solve the linear system $H_f(x_k) p_k = -\nabla f(x_k)$ for the *Newton step* p_k , then set $x_{k+1} = x_k + p_k$. Cheaper and numerically far better behaved.

Where it breaks

The step is a descent direction only when $H_f(x_k) \succ 0$. At a saddle — example 2 at the origin — Newton can move *uphill*, or fail outright when H_f is singular.

Newton on the same problem

$f(x) = 2x_1^2 + x_2^2$ from $x_0 = (1, 1)$

$$\nabla f(x_0) = \begin{pmatrix} 4 \\ 2 \end{pmatrix}, \quad H_f = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad H_f^{-1} = \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}.$$

$$x_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Exact optimum in one step.

Takeaway

No accident. If f is quadratic then $m_k = f$ exactly — the model *is* the function — so minimizing the model lands on the true minimum.

Newton solves any convex quadratic in one iteration, from any starting point.

Portfolio variance is a convex quadratic. Hold that thought.

Compare: gradient descent needed 3 steps to reach $f = 0.0012$, and would need roughly 3 more for each further factor of 10^3 .

Rates of convergence

Let $e_k = \|x_k - x^*\|$. A method converges with order p if $e_{k+1} \leq M e_k^p$ for some $M > 0$.

Order	Name	Error behaviour	Method
$p = 1, M < 1$	linear	$10^{-1}, 10^{-2}, 10^{-3}, \dots$	gradient descent
$1 < p < 2$	superlinear	faster than any fixed ratio	quasi-Newton (BFGS)
$p = 2$	quadratic	$10^{-1}, 10^{-2}, 10^{-4}, 10^{-8}, 10^{-16}$	Newton

Theorem (local quadratic convergence)

If f is C^2 , $\nabla f(x^*) = 0$, $H_f(x^*) \succ 0$ and H_f is Lipschitz near x^* , then for x_0 close enough to x^* Newton's method converges, and

$$\|x_{k+1} - x^*\| \leq M \|x_k - x^*\|^2.$$

Where it breaks

Read the words *close enough*. Quadratic convergence is a **local** guarantee. Far from the optimum, an unmodified Newton step can overshoot badly or diverge — which is why production solvers damp the step ($x_{k+1} = x_k + \alpha_k p_k$ with a line search) or modify H_f when it is not positive definite.

Gradient versus Newton — when to use which

	Gradient descent	Newton
Information used	slope only	slope and curvature
Cost per iteration	$O(n)$	$O(n^3)$ — solve a linear system
Storage	n	n^2
Iterations needed	many (linear rate)	few (quadratic rate)
On a convex quadratic	infinitely many	one
Sensitive to conditioning	very	no
Needs $H_f \succ 0$	no	yes , or must be modified

Takeaway

The practical compromise is quasi-Newton: build an approximation $B_k \approx H_f$ from successive gradients only. BFGS gets superlinear convergence at gradient-method cost, and is what sits behind most solvers you will ever call — including Excel's GRG engine.

For portfolio problems

n is the number of assets. At $n = 5$ Newton is free. At $n = 600$ a Hessian is 360,000 entries and an $O(n^3)$ solve per iteration — which is why large-scale portfolio optimizers use structure (Σ from a factor model) rather than brute force.

Where these algorithms
actually run

The same three lines, everywhere in machine learning

Every method below is $\min_{\theta} f(\theta)$ solved by $\theta_{k+1} = \theta_k + \alpha_k d_k$. Nothing in this chapter changes.

Task	What is minimized	Method
Linear / logistic regression	squared or log loss	closed form, or Newton
Neural network training	average loss over N examples	stochastic gradient descent
Clustering (k -means)	within-cluster sum of squares	alternating minimization
Feature selection	fit + sparsity penalty	coordinate / proximal descent
Support vector machine	hinge loss + $\ \cdot\ ^2$	convex QP — Chapter 3
Portfolio construction	$\mathbf{t}^T \mathbf{w} \Sigma \mathbf{w}$	convex QP — Chapter 3

Takeaway

The last two rows are the *same problem*. A support vector machine and a minimum-variance portfolio are both convex quadratic programs; only the letters differ. **What you have learned is not a finance technique — it is the engine underneath most of what is called AI.**

Neural networks: gradient descent at scale

Training minimizes the average loss over N examples, in n parameters θ :

$$\min_{\theta \in \mathbb{R}^n} f(\theta) = \frac{1}{N} \sum_{i=1}^N L(g_{\theta}(x_i), y_i).$$

Stochastic gradient descent

Computing ∇f needs all N examples. Instead estimate it on a random *mini-batch* $\mathcal{B}$:

$$\theta_{k+1} = \theta_k - \alpha_k \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \nabla L_i(\theta_k).$$

A noisy gradient step — the method from two slides ago, with the gradient sampled rather than computed.

Adam, **RMSProp** rescale each coordinate by its own running gradient magnitude: a *diagonal* correction, curvature-inspired, and cousin to the quasi-Newton idea — without ever forming H_f .

Why nobody runs Newton here

Your own table gives the answer. With $n = 10^6$ parameters — small by current standards — the Hessian has

$$n^2 = 10^{12} \text{ entries} \approx 4 \text{ TB},$$

and one Newton step costs $O(n^3) = 10^{18}$ operations.
Second-order information is simply unaffordable.

Takeaway

And f is *not convex*. Chapter 1's guarantee is gone: no stationary point is certified global. In high dimension the dominant obstacle is believed to be **saddle points**, not local minima — your $x^3 + y^3 - 3xy$ picture, in a million dimensions.

Clustering: Chapter 2 and Chapter 3 in one algorithm

k -means partitions $x_1, \dots, x_N$ into k clusters by minimizing the within-cluster sum of squares over *both* the assignment c and the centroids μ :

$$\min_{c, \mu} \sum_{i=1}^N \|x_i - \mu_{c(i)}\|^2.$$

Lloyd's algorithm — alternate

Step 1, μ fixed: assign each point to its nearest centroid. A *combinatorial* choice — Chapter 2.

Step 2, c fixed: the objective is smooth in μ_j , so set the gradient to zero:

$$\nabla_{\mu_j} = -2 \sum_{i: c(i)=j} (x_i - \mu_j) = 0 \implies \mu_j = \frac{1}{|C_j|} \sum_{i \in C_j} x_i.$$

The centroid is the mean — and that is a first-order condition from this chapter, not a definition.

Takeaway

Each step cannot increase the objective, which is bounded below, so the algorithm converges. This is **block coordinate descent**: minimize over one block of variables at a time.

Where it breaks

Solving the joint problem *exactly* is **NP-hard** — the assignment is a combinatorial object with k^N possibilities, the wall from Chapter 2. Lloyd's algorithm converges to a *local* minimum that depends on the starting centroids, which is why practitioners restart it many times (**k-means++**).

In finance: clustering assets by return correlation, before building a portfolio on cluster representatives.

Feature selection is the cardinality problem again

Choose which of p predictors to keep in a regression:

$$\min_{\beta} \|y - X\beta\|^2 \quad \text{s. t.} \quad \|\beta\|_0 \leq k \quad (\text{"use at most } k \text{ features"})$$

Best-subset selection

$\binom{p}{k}$ candidate subsets, and the constraint set is not convex — the same counterexample as the cardinality-constrained portfolio in Chapter 2. The problem is **NP-hard**.

With $p = 50$ predictors and $k = 10$: over 10^{10} subsets.

Takeaway

$\|\cdot\|_1$ is convex but not differentiable at 0, so plain gradient descent does not apply. The methods used — *coordinate descent* and *proximal gradient* — are descent methods with the non-smooth part handled in closed form. Same skeleton, one repair.

The convex relaxation — LASSO

$$\min_{\beta} \|y - X\beta\|^2 + \lambda \|\beta\|_1$$

Convex, so Chapter 1 applies: any local solution is global. And the ℓ_1 penalty drives coefficients *exactly* to zero, so it selects features as a side effect of shrinking them.

The same trade, in the client's portfolio

"Hold at most 25 names" is $\|w\|_0 \leq 25$: NP-hard.

"Penalise many small positions" is $+\lambda \|w\|_1$: a convex QP.

Feature selection and portfolio selection are the same mathematical problem, and the profession solved it the same way — by relaxing ℓ_0 to ℓ_1 .

One equality constraint

Why setting the gradient to zero is not enough

$$\min 2x_1^2 + x_2^2 \quad \text{subject to} \quad x_1 + x_2 = 1$$

Ignore the constraint for a moment and solve the unconstrained problem:

$$\frac{\partial f}{\partial x_1} = 4x_1 = 0, \quad \frac{\partial f}{\partial x_2} = 2x_2 = 0 \quad \implies \quad x_1 = x_2 = 0.$$

But $0 + 0 = 0 \neq 1$. The unconstrained minimum is infeasible, and the constrained minimum will not have $\nabla f = 0$ at all.

Takeaway

At a constrained optimum, f can still be decreasing — just not in any direction that *stays feasible*. The whole content of the next slide is to say precisely what “the leftover slope points out of the feasible set” means.

We treat only the case of a **single equality constraint**. It is all this course needs, and it is the case the portfolio problem falls into.

The Lagrangian

Method of Lagrange multipliers

For $\min f(x)$ subject to $c(x) = 0$, form

$$\mathcal{L}(x, \lambda) = f(x) - \lambda c(x)$$

and solve $\nabla_x \mathcal{L} = 0$, $\partial \mathcal{L} / \partial \lambda = 0$, i.e.

$$\nabla f(x) = \lambda \nabla c(x), \quad c(x) = 0.$$

$n + 1$ equations in $n + 1$ unknowns (x, λ) . The constrained problem has become an *unconstrained* stationarity problem in one extra variable — which is exactly why the method is worth having.

Takeaway

Geometric reading. At the optimum the gradient of the objective is *parallel* to the gradient of the constraint. If it were not, part of $-\nabla f$ would lie along the constraint surface, and we could slide along it and improve.

The same example, solved

$$\min 2x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 = 1$$

$$\mathcal{L}(x_1, x_2, \lambda) = 2x_1^2 + x_2^2 + \lambda(1 - x_1 - x_2)$$

$$\frac{\partial \mathcal{L}}{\partial x_1} = 4x_1 - \lambda = 0, \quad \frac{\partial \mathcal{L}}{\partial x_2} = 2x_2 - \lambda = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = 1 - x_1 - x_2 = 0.$$

From the first two, $x_1 = \lambda/4$ and $x_2 = \lambda/2$. Substituting into the third:

$$\frac{\lambda}{4} + \frac{\lambda}{2} = 1 \implies \frac{3\lambda}{4} = 1 \implies \lambda = \frac{4}{3}.$$

$$x^* = \left(\frac{1}{3}, \frac{2}{3}\right), \quad f(x^*) = \frac{2}{9} + \frac{4}{9} = \frac{2}{3}$$

Sanity check. $x_2^* = 2x_1^*$: the cheaper direction to move in is x_2 , whose coefficient is 2 rather than 4, so the optimum loads twice as much on it. The multiplier is what enforces the trade.

What λ means — and a bridge back to Chapter 2

Replace the right-hand side by a parameter: $\min 2x_1^2 + x_2^2$ s.t. $x_1 + x_2 = b$.

The same three equations give $\lambda = \frac{4b}{3}$, $x^* = (\frac{b}{3}, \frac{2b}{3})$ and

$$f^*(b) = \frac{2b^2}{3} \quad \implies \quad \frac{df^*}{db} = \frac{4b}{3} = \lambda.$$

Takeaway

$$\lambda = \frac{\partial(\text{optimal value})}{\partial(\text{right-hand side})}$$

The multiplier *is* the shadow price. It is the same object as the dual variable in the Chapter 2 linear program and the “Shadow Price” column in Excel’s sensitivity report — one idea wearing three names.

Where this stops. With inequality constraints $d(x) \geq 0$ the conditions acquire a sign ($\lambda \geq 0$) and a switch — complementary slackness, $\lambda_i d_i(x) = 0$: a constraint is either *active* and priced, or *slack* and free. That is the KKT system; it is named here for completeness and not developed further.

Fil rouge: the mean–variance portfolio

From a risk index to a covariance matrix

Chapter 2 maximized $\sum_i w_i \bar{r}_i$ subject to a *linear* risk index. The honest starting point is expected utility of wealth:

$$\max_w \mathbb{E} u\left(\sum_i w_i \tilde{r}_i\right) \approx \bar{R}_w - \frac{\gamma}{2} {}^t w \Sigma w, \quad \bar{R}_w = {}^t w \mu,$$

a second-order expansion of u around the mean, with γ the coefficient of risk aversion. (*Chapter 5 derives this properly.*)

One objective has become two, and there are three equivalent ways to write the problem:

Form	Problem	Chapter
Utility	$\max {}^t w \mu - \frac{\gamma}{2} {}^t w \Sigma w$	3, 5
ε -constraint	$\max {}^t w \mu \text{ s.t. } {}^t w \Sigma w \leq \varepsilon$	4
Markowitz	$\min \frac{1}{2} {}^t w \Sigma w \text{ s.t. } {}^t w \mu \geq R_0, {}^t 1 w = 1$	3

Takeaway

All three trace the same *efficient frontier*; they differ only in which knob the client turns. Chapter 4 shows this is a general fact about multi-objective problems, not a coincidence about portfolios.

The one-constraint case, in closed form

Drop the return target and keep only the budget — **a single equality constraint**, exactly the case we solved two slides ago:

$$\min_w \frac{1}{2} {}^t w \Sigma w \quad \text{s. t.} \quad {}^t 1 w = 1.$$

$$\begin{aligned} \mathcal{L}(w, \lambda) &= \frac{1}{2} {}^t w \Sigma w + \lambda(1 - {}^t 1 w) \\ \nabla_w \mathcal{L} = \Sigma w - \lambda 1 &= 0 \implies w = \lambda \Sigma^{-1} 1. \end{aligned}$$

Impose ${}^t 1 w = 1$: $\lambda {}^t 1 \Sigma^{-1} 1 = 1$, so $\lambda = 1/({}^t 1 \Sigma^{-1} 1)$ and

$$w^* = \frac{\Sigma^{-1} 1}{{}^t 1 \Sigma^{-1} 1} \quad \text{the global minimum-variance portfolio.}$$

Takeaway

Three lines of Lagrange give a formula every practitioner knows. Note the by-products: $\Sigma \succ 0$ makes the objective *strictly convex*, so by Chapter 1 the solution is **unique and global**; and the portfolio variance at the optimum is exactly $\lambda = 1/({}^t 1 \Sigma^{-1} 1)$.

NEOMAMA, with a covariance matrix

Fil rouge — the client's three live positions

Chapter 2's optimum used only municipal bonds, Palmer Drugs and the nursing home stock. Estimate their covariances and re-solve.

	$\bar{r}_i$	σ_i
Municipal bonds	5.3%	2%
Palmer Drugs	8.4%	5%
Nursing Homes	11.8%	8%

$$\Sigma = \begin{pmatrix} 4 & 2 & 1 \\ 2 & 25 & 12 \\ 1 & 12 & 64 \end{pmatrix} (\%^2)$$

correlations $\rho_{12} = 0.20$, $\rho_{13} = 0.06$, $\rho_{23} = 0.30$.

Minimum-variance portfolio

$$\mathbf{w}^* = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^T \Sigma^{-1} \mathbf{1}} = \begin{pmatrix} 0.904 \\ 0.063 \\ 0.033 \end{pmatrix}$$

Return 5.71%, standard deviation 1.94% — below the 2% of the safest single asset.

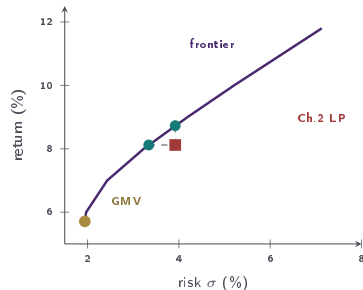
That is diversification, quantified. No linear model can produce a portfolio less risky than every asset in it. The cross terms in $\mathbf{w}^T \Sigma \mathbf{w}$ are the entire reason it happens.

What the linear model cost her

	Bonds	Palmer	Nursing
Chapter 2 (LP)	20%	70%	10%
Same return, min variance	40%	31%	29%

	Return	Risk σ
Chapter 2 (LP)	8.12%	3.92%
Same return, min variance	8.12%	3.34%
Same risk, max return	8.73%	3.92%

Two ways to state the same gain: **15% less risk** for the same return, or **+61 basis points** of return for the same risk.



Where it breaks

The Chapter 2 answer sits *inside* the frontier — it is not inefficient because the LP was solved badly, but because the model had no covariance in it. **The error was in the formulation, not the solver.**

Workshop — the same problem in Excel

Sheet layout

- B3:D3 — weights w (changing cells)
- B5:D5 — expected returns μ
- B7:D9 — covariance matrix Σ
- portfolio return
`=SUMPRODUCT(B3:D3, B5:D5)`
- portfolio variance
`=MMULT(MMULT(B3:D3, B7:D9),
TRANSPOSE(B3:D3))`
- budget `=SUM(B3:D3) = 1`

Solver

Set Objective: portfolio variance, **To:** Min

Changing cells: B3:D3

Subject to: `SUM(B3:D3) = 1`

and, for the frontier, `return >= R0`

Solving Method: **GRG Nonlinear**

Where it breaks

Simplex LP will refuse this model — the objective is not linear. GRG Nonlinear is a gradient method with a quasi-Newton engine: everything in the first half of this session, running under a button.

Tasks. (1) Reproduce $w^* = (0.904, 0.063, 0.033)$ and $\sigma = 1.94\%$. (2) Add `return >= 8.12` and confirm $\sigma = 3.34\%$. (3) Sweep R_0 from 6% to 11% and plot the frontier. (4) Tick *Make Unconstrained Variables Non-Negative* and re-run at $R_0 = 11.5\%$ — explain what changes and why.

Problems

Problems — set A: derivatives and optimality

1. Let $f(x_1, x_2) = (x_1 + x_2^2)^2$. Derive the gradient and the Hessian at $(1, 2)$. Is $H_f(1, 2)$ positive definite? Is $(1, 2)$ a local minimum? Answer both, and say why they are different questions.
2. Let $f(x, y) = xy^2 + x^2y - xy$. Compute ∇f and H_f at $(1, 1)$. Find all stationary points and classify each one.
3. Let $f(x, y) = x^3 + y^3 - 3xy$. Confirm the two stationary points and their classification. Then show directly that f decreases along $y = x$ through the origin and increases along $y = -x$.
4. Expand $f(x) = \ln(1 + x)$ to second order about $x_0 = 0$. Use it to approximate $\ln(1.1)$, and compare with the true value. Now do the same for $\ln(2)$ and comment.
5. Let $f(w) = w^T \Sigma w$ with Σ symmetric. Show that $\nabla f = 2\Sigma w$ and $H_f = 2\Sigma$. Deduce that f is convex whenever Σ is a covariance matrix, and strictly convex when $\Sigma \succ 0$.

Problems — set B: algorithms and the portfolio

6. **By hand.** For $f(x) = 2x_1^2 + x_2^2$ from $x_0 = (2, -1)$: (a) two steps of gradient descent with exact line search; (b) one step of Newton. Comment.
7. **Conditioning.** Repeat problem 6 for $f(x) = 50x_1^2 + x_2^2$ from $(1, 1)$. How many gradient steps to reach $f < 10^{-3}$? How many Newton steps? Relate your answer to $\kappa = \lambda_{\max}/\lambda_{\min}$.
8. **Lagrange.** Solve $\min x_1^2 + 2x_2^2 + 3x_3^2$ subject to $x_1 + x_2 + x_3 = 6$. Give x^* and λ . Then verify numerically that $\lambda = df^*/db$ by re-solving with right-hand side 6.1.
9. **Minimum variance.** Two assets, $\sigma_1 = 10\%$, $\sigma_2 = 20\%$, correlation ρ . Show that $w_1^* = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$. Evaluate at $\rho = -1, 0, +1$ and interpret each.
10. **Fil rouge.** Using the NEOMAMA covariance matrix: (a) reproduce w^* from the closed form; (b) compute the portfolio at $R_0 = 8.12\%$ in Solver; (c) state, in one sentence for the client, what the Chapter 2 allocation cost her.
11. **Synthesis.** The Chapter 2 LP and the Chapter 3 QP were given the same data and the same client. Explain why the QP answer is better without using the words “optimal” or “efficient”.

Summary

Theory

- Taylor: near a point, every C^2 function is a parabola.
- ∇f gives the slope, H_f the curvature.
- Necessary: $\nabla f(x_0) = 0$, $H_f(x_0) \succeq 0$.
Sufficient: $\nabla f(x_0) = 0$, $H_f(x_0) \succ 0$.
- $\det H < 0$ in two variables means a saddle.
- Convex f : every stationary point is global (Ch.1).
- One equality constraint: $\nabla f = \lambda \nabla c$, and λ is the shadow price (Ch.2).

Algorithms and application

- $x_{k+1} = x_k + \alpha_k d_k$ is the shape of all of them.
- Gradient descent: cheap, linear rate, zig-zags when κ is large.
- Newton: $x_{k+1} = x_k - H_f^{-1} \nabla f$, quadratic rate, one step on a quadratic, local guarantee only.
- Quasi-Newton is the working compromise — and is what GRG Nonlinear runs.
- $w^* = \Sigma^{-1} \mathbf{1} / (\mathbf{1}^T \Sigma^{-1} \mathbf{1})$: three lines of Lagrange.
- Diversification is the cross terms of $\mathbf{1}^T w \Sigma w$, and nothing else.

Takeaway

The client now has an efficient portfolio for *one* risk tolerance. She has not told us what her risk tolerance is — and she also cares about things the covariance matrix does not contain. **Chapter 4** stops collapsing return and risk into one number and treats them as what they are: genuinely separate objectives.

Reading. Full references, with what each source is for, on the next slide.

References

References and further reading

Core

- Nocedal, J. & Wright, S. J. (2006). *Numerical Optimization*, 2nd ed. Springer. ch. 2–6: line search, Newton and quasi-Newton, convergence rates. [The reference for this chapter.](#)
- Boyd, S. & Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press. ch. 9 for descent methods, ch. 5 for Lagrange multipliers and KKT.

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- Markowitz, H. (1952). Portfolio selection. *J. Finance* 7(1), 77–91. [seven pages; worth reading in the original.](#)
- Cornuéjols, G., Peña, J. & Tütüncü, R. (2018). *Optimization Methods in Finance*, 2nd ed. Cambridge University Press. [quadratic programming in portfolio construction.](#)
- Hastie, T., Tibshirani, R. & Friedman, J. (2009). *The Elements of Statistical Learning*, 2nd ed. Springer. ch. 3–4: the LASSO, logistic regression and IRLS — [the descent methods of this chapter, at work.](#)

Takeaway

Next. Chapter 4 — multi-objective optimization: Pareto dominance, the weighted sum, the ϵ -constraint method, and the efficient frontier as a Pareto front.